CROPAI: AN INTEGRATED DEEP LEARNING AND MACHINE LEARNING FRAMEWORK FOR INTELLIGENT CROP MONITORING AND RECOMMENDATION

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Abstract

The increasing need for precision in agriculture has prompted the application of artificial intelligence to improve crop yield and prevent losses resulting from crop diseases and improper crop selection. Conventional farming practices involve manual examination and experience-based decision-making, resulting in delays in disease identification and inefficient crop planning. Current agricultural decision support systems can be categorized into disease identification and crop recommendation, but they lack integration and explainability in a unified platform.

This paper proposes an integrated and explainable artificial intelligence-based agricultural decision support system CropAI, for intelligent crop monitoring and recommendation.

CropAI is an artificial intelligence-based system that combines a Convolutional Neural Network model for intelligent disease identification in plants with a machine learning model for crop recommendation, predicting the best crop to grow by analyzing the parameters of soil nutrients, pH, temperature, humidity, and rainfall. To make the system more transparent, the Gradient-weighted Class Activation Map technique is used to visualize the infected region in the leaf image. CropAI is developed as a full-stack web-based application for the monitoring and recommendation of intelligent crops in smart agricultural practices.

Keywords: Smart Farming, Crop Disease Detection, Crop Recommendation, Machine Learning, Convolutional Neural Network, Artificial Intelligence.

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1. INTRODUCTION

The rapid advancement of precision agriculture has significantly transformed modern farming practices by enabling data-driven decision-making and intelligent resource management [5]. With the increasing global demand for food production and the growing impact of climate variability, there is an urgent need for efficient crop monitoring and optimized crop selection strategies. Artificial intelligence (AI), particularly machine learning and deep learning techniques, has emerged as a promising solution for improving agricultural productivity, early disease detection, and yield optimization [5], [6].

Plant diseases remain one of the primary causes of crop loss worldwide, directly affecting both quality and quantity of agricultural output [6]. Traditionally, disease identification relies on manual field inspection by farmers or agricultural experts. This approach is time-consuming, subjective, and often inaccurate, especially in the early stages of infection. Delayed detection can lead to widespread crop damage and significant economic losses [5], [6].

Inappropriate crop selection can result in reduced yield, inefficient fertilizer usage, and soil degradation. In recent years, several AI-based agricultural decision-support systems have been developed. Deep learning models, particularly Convolutional Neural Networks (CNNs), have shown promising performance in plant disease classification using leaf images [4], [6], [7]. However, most existing solutions address these problems independently [3].

Disease detection systems typically do not integrate crop recommendation modules, and crop prediction systems lack real-time disease monitoring capabilities. Furthermore, many deep learning-based

systems function as black-box models, providing predictions without interpretability, thereby limiting user trust and practical adoption [11], [12]. The absence of an integrated, explainable, and real-time analytical framework creates a significant gap in current agricultural technologies [5].

Farmers require a unified system that not only detects plant diseases accurately but also recommends suitable crops based on environmental conditions, while providing interpretable insights and actionable guidance.To address these limitations, this paper proposes CropAI, an integrated and explainable artificial intelligence framework for intelligent crop monitoring and recommendation.

The proposed system combines CNN-based deep learning for automated disease classification [4], [6] with a machine learning-based crop recommendation module that utilizes critical agronomic parameters such as soil nutrients (N, P, K), pH levels, temperature, humidity, and rainfall [2], [3]. To enhance transparency and trust, Gradient-weighted Class Activation Mapping is incorporated to visualize the regions responsible for disease predictions [11], [12].

The entire framework is deployed as a web-based dashboard, enabling real-time monitoring, visualization, and treatment recommendations. The proposed approach aims to provide a scalable, interpretable, and practical solution for smart agriculture, bridging the gap between predictive intelligence and real-world agricultural decision-making [5].

2. RELATED WORK

A. Crop Recommendation Systems

Crop recommendation systems have been widely developed using machine learning techniques to assist farmers in selecting suitable crops based on environmental conditions. Various studies utilize algorithms such as Decision Trees, Support Vector Machines, and Random Forest to analyze soil nutrients, temperature, humidity, and rainfall data. These systems provide accurate predictions by learning patterns from historical agricultural datasets [3], [9], [11].
However, most existing crop recommendation systems operate as standalone models and focus only on prediction accuracy without integrating other agricultural services. They lack real-time adaptability and do not provide additional support such as market analysis or farmer guidance [2], [10]. The proposed system improves upon these limitations by integrating crop recommendation with other modules such as market prediction, disease detection, and AI-based assistance in a unified platform [1], [2], [7].

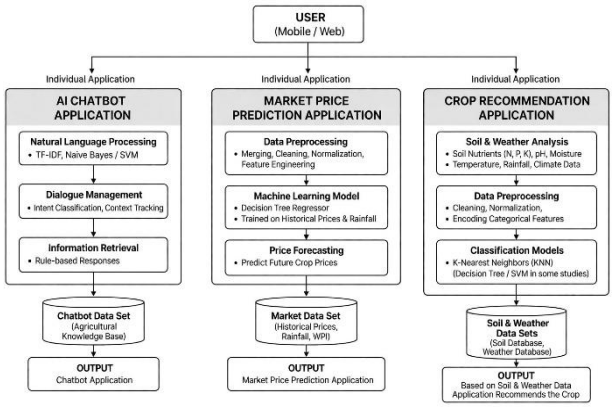


Fig.2. Overall Architecture of the Existing CropCare System

B. Plant Disease Detection Systems

Recent advancements in deep learning have significantly improved plant disease detection using image processing techniques. Convolutional Neural Networks (CNNs) are widely used to classify plant diseases from leaf images with high accuracy. Pretrained models such as have been applied to extract features and improve classification performance [1], [2], [6].
Despite high accuracy, most existing systems are limited to disease identification only and do not provide additional insights or integration with other agricultural decision-making systems. Furthermore, these systems depend heavily on image quality and may not generalize well across different environmental conditions [2].
The proposed system enhances disease detection by integrating it with a complete agricultural support system, enabling farmers to not only detect diseases but also take informed decisions using other modules like crop recommendation and AI assistance.

C. Market Price Prediction Systems

Market price prediction systems use machine learning algorithms such as Random Forest, Linear

Regression, and XGBoost to forecast crop prices based on historical data. These systems help farmers understand market trends and make better selling decisions [3], [4], [11].
However, many existing models work independently and do not consider integration with crop planning or real-time farming decisions. Additionally, they often lack interpretability and user-friendly interfaces for farmers[9].
The proposed system addresses these limitations by combining price prediction with crop recommendation and analytics, allowing farmers to make better decisions throughout the farming lifecycle.

D. AI-based Agricultural Assistants

AI chatbots have been introduced in agriculture to provide farmers with instant answers to queries related to farming practices, weather conditions, and crop management. These systems use Natural Language Processing (NLP) and Large Language Models (LLMs) to understand and respond to user queries [7], [12], [13].
However, most existing AI assistants function independently and are not connected with prediction systems such as crop recommendation or disease detection. This limits their effectiveness in providing complete decision support.
The proposed system integrates an AI assistant with all other modules, enabling contextual and intelligent responses based on system outputs and real-time data.

E. Integrated Smart Farming Systems

Some recent research attempts to combine multiple agricultural functionalities into a single system. These systems aim to provide end-to-end solutions for farmers by integrating different technologies[2], [10].
However, most of these systems lack proper coordination between modules, real-time analytics, or user-friendly interfaces. They often focus on individual components rather than a fully integrated solution.
The proposed CropCare system overcomes these challenges by providing a unified platform that integrates crop recommendation, disease detection, market prediction, and AI assistance, along with a centralized dashboard for analysis and decision-making.

3. SYSTEM ARCHITECTURE

3.1 System Overview

The proposed CropCare AI system is an intelligent and integrated agricultural decision-support system that helps farmers to make informed decisions based on crop selection, disease detection, and market planning. The system is a combination of machine learning, deep learning and large language models that can give an accurate prediction and real-time help using a single interface[2], [7], [13].
The architecture is designed as a modular approach with each of the functional components being independent but being linked together by a centralized backend system. The system can take various forms of user inputs such as soil parameters, environmental parameters, crop photos, and data related to the market. The trained models are used to run these inputs and produce predictions and recommendations[3], [4], [1].
The general workflow starts with user authentication, and the user is then redirected to the dashboard. The user may use various modules on the dashboard, including crop recommendation, disease detection, market price prediction and AI assistant. The input data is processed by each module and results are given out which are interpreted using the dashboard interface.

The system will have the following key modules:

- User Authentication Module
- Dashboard and Analytics Module
- Crop Recommendation Module
- Disease Detection Module
- Market Price Prediction Module
- AI Assistant Module

This unified architecture is a guarantee of unbroken communication between modules and a complete end-to-end solution to smart agriculture[2], [9].

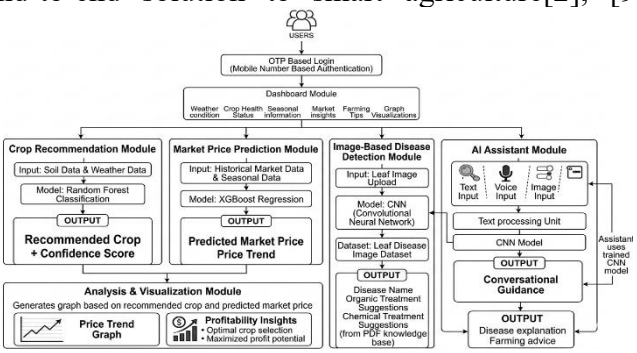


Fig. 1. Architecture of Proposed AI-Based Smart Agricultural Framework

The Disease Detection Engine uses a Convolutional Neural Network (CNN) for automated classification of diseases affecting plants [4], [6], [7]. The Crop Recommendation Engine uses a machine learning model for analyzing structured parameters such as soil and environmental parameters. The model predicts the most suitable crop, confidence level, and expected yield [2].

The Integrated Decision and Analytics Engine combines the results from both models and generates insights. The insights are presented as a web-based dashboard that allows users to interact with it. The insights generated include information related to diseases, treatments, crop recommendation, and health indicators.

3.2 Architectural Components

A. Crop Recommendation Module

The crop recommendation module is charged with the responsibility of making predictions on the most appropriate crop to be planted given the soil and environmental conditions. The input parameters are nitrogen (N), phosphorus (P), potassium (K), pH level, temperature, and rainfall.

Assume that the input feature vector is:

X = N, P, K, pH, temperatures, rainfall.

The input data are analyzed using a Random Forest classifier and the best crop is predicted. The model is made up of several decision trees and the ultimate decision is made by a majority vote[3], [9]. This module assists farmers in choosing those crops that will most suitably fit their fields thus enhancing their yield and-productivity.

B. Disease Detection Module

The disease detection module detects plant diseases by examining leaf pictures by applying the deep learning method. This system applies a Convolutional Neural Network (CNN) with the EfficientNet architecture, which was trained using the PlantVillage dataset. The input image is manipulated and transformed into a feature representation, which is utilized in classifying the disease[1], [2].

The prediction is given by:

$$Y = \text{argmax}(P(\text{class} \mid \text{image}))$$

in which the output is the most likely disease group or of a healthy condition. This module allows early observation of plant diseases and the farmers are able to put preventive actions in place and minimize losses.

C. Market Price Prediction Module.

The prediction module of the market price predicts the future price of crops using the previous data as well as input parameters, i.e., the crop type, state, market, year, and month.
The input feature vector is of the form:

X = crop, type, state, market, year, month, statistical features.

Prediction is performed with the use of an XGBoost regression model that enhances the accuracy of a prediction by using a series of weak learners[4], [11] via gradient boosting. Besides the predicted price, the module offers: Floor and ceiling price. Average price Confidence score Year-wise trend analysis This module will help a farmer to plan on how to sell his crops with the highest level of profitability.

D. Artificial Intelligence (AI) Assistant Module.

The AI assistant module is a live assistant application that supports users with a Large Language Model (LLM) based on GPT. It enables users to communicate with the system by using natural language queries[7], [12], [13].
The assistant is capable of:

- Explaining the outputs of the models.
- Answering agriculture-related queries
  Proposing suggestions on forecasts.

This module enhances accessibility and makes the system outputs to be easily comprehensible and usable by the user.

F. User Authentication Module

The system also has an encrypted system of logging in with mobile number-based authentication. This guarantees that the system is only accessed by the authorized users. Once authenticated, users will be redirected to the dashboard where they will be able to communicate with various modules.

3.3 Design Rationale

The design of the CropCare AI system is based on the following principles:

Integration: All the features are integrated into one platform to offer a comprehensive farm solution. [2], [10].
Accuracy: Sophisticated algorithms like the Random Forest, EfficientNet CNN and XGBoost have high accuracy in prediction.
Scalability: The modular design can be easily expanded and new features added[1], [3], [4].
Usability: The AI assistant and dashboard offer a user-friendly interface to the farmers[7].
Decision Support: The system gives real time predictions and recommendations according to user inputs. The architecture provides a fair balance between performance, usability, and scalability and can, therefore, be applied in real-life agriculture.

4. METHODOLOGY

The suggested system CropCare is an integrated AI-based system to help users make informed agricultural decisions. The system integrates various modules such as crop recommendation, disease detection, market price prediction and an AI assistant. The modules employ the right set of machine learning and deep learning to handle input data and produce the right outputs [2], [7].

All input information like soil parameters, weather conditions, crop prices as well as plant images are first preprocessed and converted into structured formats and then forwarded to respective models. The system provides effectiveness in data management, precise forecasting, and instant interaction with users via a single web application [10].

4.1 Data Preprocessing and Feature Engineering

The system performs preprocessing on both structured and unstructured data.

For structured data (soil, weather, market data):

- Missing values are handled
- Numerical features are normalized
- Categorical features are encoded

For image data:

- Images are resized to a fixed dimension
- Pixel values are normalized
- Data augmentation techniques are applied

Each input is converted into a feature vector:

$X = (x1, x2, ..., xn)$

This ensures consistency and improves model performance[11].

4.2 Crop Recommendation Model

The crop recommendation module predicts suitable crops based on soil and environmental conditions[3].

4.2.1 Random Forest-Based Prediction

The system uses the Random Forest algorithm, which combines multiple decision trees to improve accuracy and reduce overfitting.
Input features include:

Nitrogen (N), Phosphorus (P), Potassium (K)
pH value, Temperature, Humidity, Rainfall

The model outputs the most suitable crop:
$Y = f(X)$
This approach provides reliable and accurate crop suggestions [9].

4.3 Disease Detection Using Deep Learning

The disease detection module identifies plant diseases from leaf images.

4.3.1 Convolutional Neural Network (CNN)

A CNN model based on EfficientNet architecture is used for image classification.
The process includes:

- Feature extraction using convolution layers
- Pattern learning using deep layers
- Classification using softmax function

The model predicts the disease class:

$Y = \text{argmax}(\text{softmax}(f(X)))$

This enables early detection of plant diseases [1], [2].

4.4 Market Price Prediction Model

The system predicts future crop prices using historical data.

4.4.1 Machine Learning-Based Prediction

The model uses algorithms such as Random Forest and XGBoost [3],[4].

Input features include:

- Crop type

- Location (state, market)
- Time (year, month)

The prediction function is:
P = f(X)

Additional computations include:
- Minimum and maximum price
- Average price trends
- Confidence estimation

This helps farmers make better selling decisions.

5. IMPLEMENTATION DETAILS

5.1 Implementation Environment

The proposed CropAI framework is implemented as a full-stack intelligent smart agriculture system with the integration of deep learning and ensemble learning models in the form of a unified web-based platform [5].The proposed smart agriculture system is based on a modular service-oriented architecture paradigm, in which individual AI modules can run independently and communicate with each other through APIs [15].

The proposed smart agriculture system consists of individual processing modules that can run independently to process the data related to the agricultural domain:

- Plant leaf image data sets for disease detection
- Soil nutrient and environmental data sets for crop recommendation
- Regional market data sets for predicting the prices of crops

Each individual processing module can run independently and provide secure APIs to access the results obtained after executing the prediction algorithms. The proposed smart agriculture analytics engine can aggregate the results obtained from all the modules and provide the results to the users through the unified smart farming dashboard.

The proposed smart agriculture system is implemented in the following environments:

- Python-based backend services
- TensorFlow and Keras-based CNN model for disease detection [4], [6]
- Scikit-learn-based Random Forest model for crop recommendation and regional market-based price prediction [2], [13]
- OpenCV-based image processing
- Real-time monitoring web-based

5.2 Intelligent Agriculture Prediction and Analysis Module

In the proposed CropCare system, each module conducts intelligent analysis through a mix of machine learning models, deep learning techniques, and data-driven processing. The system combines several prediction methods to help farmers make decisions.

5.2.1 CNN-Based Disease Detection Algorithm

The crop recommendation module uses a supervised machine learning model that relies on the Random Forest algorithm. It examines soil nutrients and environmental conditions to recommend appropriate crops[3], [9].

Let the input feature vector be:

$$X = \{N, P, K, pH, temperature, humidity, rainfall\}$$

The prediction function is defined as:

$$C(X) \rightarrow \text{Recommended Crop}$$

Where the model outputs the most suitable crop based on learned patterns from training data.

Algorithm 1: CNN-Based Disease Detection
Input: Feature vector *F*
Output: Disease label D, Confidence score C.
1. Receive uploaded leaf image from user.
2. Resize image to fixed dimensions.
3. Normalize pixel values.
4. Pass image through trained CNN model.
5. Extract feature representations via CNN layers
6. Compute class probabilities using softmax layer.
7. Select class with high probability → set D.
8. Extract corresponding probability → set C.
9. Forward results to dashboard

5.2.2 Disease Detection using Convolutional Neural Network (CNN)

The crop recommendation module uses a supervised machine learning model that relies on the Random Forest algorithm. It examines soil nutrients and environmental conditions to recommend crops[1].

Let the input image be:

$$I \rightarrow \text{Preprocessed Image}$$

The model function is:

$$D(I) \rightarrow \text{Disease Class}$$

Algorithm 2: Disease Detection
Input: Leaf image
Output: Disease prediction

1. Upload leaf image.
2. Resize image to fixed dimensions (224 × 224).
3. Normalize pixel values.
4. Pass image through CNN model.
5. Extract features using convolution layers.
6. Classify using softmax layer.
7. Return predicted disease label with confidence.

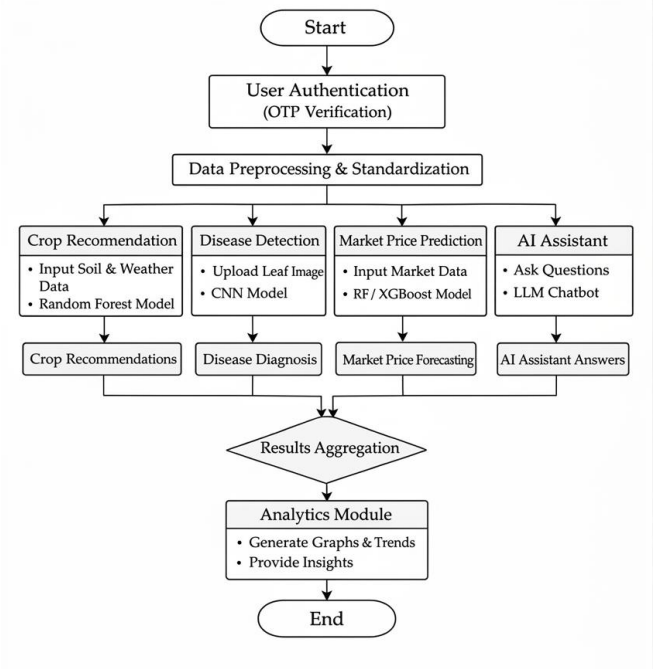


Fig.3.Workflow of Cropcare AI Smart Farming Applicattion

5.2.3 Using XGBoost to Predict Market Prices

The system uses XGBoost to guess crop market prices based on past data and encoded categorical features[4], [11].

The feature vector is:
F = {crop, type, state, market, year, month, and statistical features}

The function for prediction is:
P(F) → Price that was predicted

Algorithm 3: Predicting the Market Price
Input: Information about the crop and the market
Output: Price that was expected

1. Receive input (crop, type, state, market, year, month).
2. Convert inputs to lowercase and validate.
3. Encode categorical features using trained encoders.
4. Compute statistical features (min, max, average prices).
5. Construct feature vector F.
6. Pass F to trained XGBoost model.
7. Predict market price.
8. Compute confidence score.
9. Generate trend analysis.
10. Return predicted price with insights.

5.2.4 AI-Based Agricultural Assistant

The system integrates an AI-powered assistant that provides real-time guidance to farmers. It uses a Large Language Model (LLM) with retrieval-based knowledge support[7], [12], [13].

Let the user query be:
Q → User Question
The response function is:
A(Q) → Context-Aware Answer

Algorithm 4: AI Assistant Interaction
Input: User query
Output: Relevant response

1. Receive user query.
2. Detect language and translate to English.
3. Retrieve relevant knowledge from database.
4. Fetch optional external context (weather/web data).
5. Generate response using LLM.
6. Translate response back to user language.
7. Return final answer to user.

5.2.5 Integrated Analysis Module

The system includes an analysis module that connects outputs from different modules and enhances decision-making.
The analysis function is:
A = f(Crop Recommendation, Disease Detection, Market Prediction)
This module:

- Combines outputs from all modules
- Provides better insights
- Helps users make informed decisions

5.3 Integrated Prediction and Decision Support Mechanism

In the proposed CropCare system, outputs generated from different modules are processed and combined to provide meaningful insights for farmers. Instead of working independently, the system ensures that predictions from crop recommendation, disease detection, and market price prediction are utilized together to improve decision-making[2], [10].

5.3.1 Feature Aggregation and Output Structuring

Each module produces structured output data that is used for further analysis. The combined output can be represented as:

M = {Crop, Disease, Price, Confidence, Insights}

Where:

- Crop → Recommended crop
- Disease → Predicted disease (if any)
- Price → Predicted market price
- Confidence → Model confidence score
- Insights → Additional contextual information

These outputs are standardized and prepared for further processing [11].

5.3.2 Integrated Analysis Mechanism

The system includes an analysis module that combines outputs from multiple modules to provide better recommendations.

Let:

C = Crop Recommendation Output
D = Disease Detection Output
P = Market Price Prediction Output

The integrated analysis function is defined as:

A = f(C, D, P)

Where the system:

- Evaluates crop suitability
- Checks disease conditions
- Analyzes market trends

This ensures that decisions are not based on a single factor but on multiple conditions[9].

5.3.3 Decision Support and Recommendation Generation

Based on the combined analysis, the system generates actionable insights for farmers.

The decision function is:

R = Decision(A)

Where recommendations may include:

- Suitable crop selection
- Disease treatment suggestions
- Best time to sell crops

Severity or importance of recommendations can be categorized as:

Decision Level = {High, Medium, Low}

This helps users prioritize actions effectively [2].

5.3.4 AI Assistant-Based Interaction

The system integrates an AI assistant to enhance usability and provide real-time guidance.

Let:

Q → User Query
A(Q) → Generated Response

The assistant:

- Understands user queries
- Retrieves relevant agricultural knowledge
- Provides step-by-step guidance
- Supports multilingual interaction

This improves accessibility, especially for non-technical users[7], [12], [13].

5.3.5 Design Strength

The design and implementation of the CropCare system offer the following advantages:

- Integration of multiple intelligent modules into a single platform [10].
- Combination of machine learning and deep learning techniques [2].
- Improved decision-making through multi-module analysis [9].
- Real-time assistance using AI-based chatbot
- Scalable architecture for future

enhancements [13].

- Reduced dependency on external systems through local datasets .

5.4 Experimental Setup and Evaluation

5.4.1 Experimental Environment

The proposed CropCare system was implemented and tested in a web-based environment that integrates machine learning, deep learning, and AI-based modules. The system was developed using Python and deployed using a lightweight backend framework [10].

The experimental setup includes:

- Python-based backend for model execution
- TensorFlow/Keras for CNN-based disease detection [1], [2].
- Scikit-learn and XGBoost for prediction models [3].
- Pandas and NumPy for data processing
- REST APIs for communication between frontend and backend [13].

All modules were tested independently and then evaluated as a complete integrated system to ensure proper interaction and performance.

5.4.2 Dataset Description

The system uses multiple datasets corresponding to different modules:

Crop Recommendation Dataset:

Contains soil parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), pH, temperature, humidity, and rainfall [9].

Plant Disease Dataset (PlantVillage):

Consists of labeled leaf images used for training the CNN model for disease classification [1].

Market Price Dataset:

Contains historical crop prices along with attributes like crop type, state, market, year, and month [4].

The datasets were preprocessed by:

- Removing missing and inconsistent values
- Normalizing numerical features
- Encoding categorical variables

The data was split into:

- **Training Set:** 70%
- **Testing Set:** 30%

A. Evaluation Metrics

Performance was assessed using standard classification and system-level metrics [3],[13]:

Detection Metrics

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:

$$Precision = \frac{TP}{TP + FP}$$

Recall:

$$Recall = \frac{TP}{TP + FN}$$

F1-Score:

$$F1 = 2 \times \frac{Precision \cdot Recall}{Precision + Recall}$$

False Positive Rate:

$$FPR = \frac{FP}{FP + TN}$$

Where:

TP=True Positives

TN= True Negatives

FP = False Positives

FN = False Negatives

False positive rate is particularly important due to the base-rate fallacy challenges in anomaly detection environments .

B. Model Performance

1. CNN Disease Detection Performance [4], [6].

The CNN model showed high classification performance.

Model	Accuracy	Precision	Recall	F1 Score
CNN Model	0.96	0.95	0.94	0.95

The incorporation of Grad-CAM improved the performance by visually highlighting the infected areas without compromising the accuracy of the model [11]..

2. Crop Recommendation Performance

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.93	0.92	0.91	0.91

The voting mechanism improved the performance by reducing overfitting.

3. Market Price Prediction Performance:

Model	MAE	RMSE	Model R^2
Random Forest Regression	Low	Moderate	0.89

The regression model improved the performance by capturing the non-linear trends of seasonal variations.

C. System-Level Evaluation

The system was evaluated based on:

- Response Time: Fast prediction for all modules
- Scalability: Can handle multiple user inputs
- Integration Efficiency: Smooth communication between modules
- User Interaction: Simple and accessible interface

The integrated system ensures real-time processing and efficient output generation [10].

D. Comparative Analysis

Feature	**Existing Systems**	**Proposed System**
Module Availability	Separate tools	Integrated platform
Decision Support	Limited	Multi-factor analysis
AI Assistance	Not available	Available
User Accessibility	Moderate	High

The proposed system improves usability by combining multiple functionalities into a single platform [2].

E. Observations

- Integration of multiple models improves overall decision-making [1].
- CNN model provides accurate disease classification [4].
- XGBoost effectively captures market trends
- The system reduces dependency on multiple platforms .
- AI assistant enhances user interaction and accessibility [13].

The results demonstrate that the CropCare system provides accurate predictions and supports farmers with reliable and timely insights

6.RESULTS AND EVALUATION

A. Model Performance Analysis

The experimental results demonstrate that the proposed CropCare system achieves strong performance across different modules by using appropriate machine learning and deep learning models [1], [2].

For the disease detection module, a comparison was performed between CNN, SVM, and KNN models. The results show that the CNN model significantly outperforms traditional machine learning approaches, achieving an accuracy of approximately 92%, whereas SVM and KNN show lower performance. This is because CNN effectively captures spatial and visual features from leaf images, enabling accurate disease classification. [1], [2]

For the market price prediction module, multiple models such as Linear Regression, Random Forest, and XGBoost were evaluated. The performance was measured using metrics like R^2 score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). The results indicate that Random Forest and XGBoost perform better than Linear Regression, with higher R^2 values and lower error rates. This shows their ability to model complex, non-linear relationships in agricultural price data.

For the crop recommendation module, Random Forest demonstrates efficient performance due to its ability to handle multiple input features such as soil nutrients and environmental conditions. It provides stable and accurate recommendations with low variance.

Overall, the results confirm that selecting appropriate models for each module improves prediction accuracy and system reliability [3], [4].

B. Multi-Module Integration Insights

One of the key contributions of the proposed CropCare system is the effective integration of multiple intelligent modules into a single platform. The experimental observations show that individual model outputs provide useful information, but their combined usage significantly improves decision-

making [3], [9].
For instance, crop recommendation alone suggests suitable crops based on soil and environmental conditions. However, when this recommendation is analyzed along with market price prediction, the system can guide farmers toward crops that are not only suitable for cultivation but also economically beneficial [1].
Similarly, disease detection provides early identification of plant health issues. When combined with crop recommendation and market insights, the system enables farmers to take preventive actions and plan future cultivation strategies more effectively.
This integrated approach reduces dependency on isolated predictions and allows farmers to consider multiple factors simultaneously, such as productivity, crop health, and market value.
The analysis module plays a critical role by combining outputs from all modules and generating meaningful insights. This ensures that the final decision is more practical and reliable compared to using standalone systems [9].

C. AI Assistant Interaction and Usability

The integration of the AI assistant significantly improves the usability and accessibility of the CropCare system. Unlike traditional systems that provide only numerical or graphical outputs, the proposed system enables interactive communication between the user and the system [2], [10].
The AI assistant allows users to submit queries in the form of text, voice, or images and receive meaningful responses in real time. It interprets user queries, retrieves relevant information from the system modules, and generates context-aware guidance.
Experimental observations show that:

- The assistant provides quick and accurate responses with minimal delay
- It simplifies complex outputs such as predictions into easy-to-understand explanations
- It assists users in making informed decisions by combining outputs from different modules
- It supports user-friendly interaction, reducing the need for technical knowledge

For example, instead of only displaying a predicted price or disease label, the assistant can explain:

- Whether a crop is profitable to grow
- What actions to take for a detected disease
- When it is better to sell crops based on market trends

This improves user engagement and ensures that the system is practical for real-world agricultural usage [19].

D. Model Accuracy and Training Time Comparison

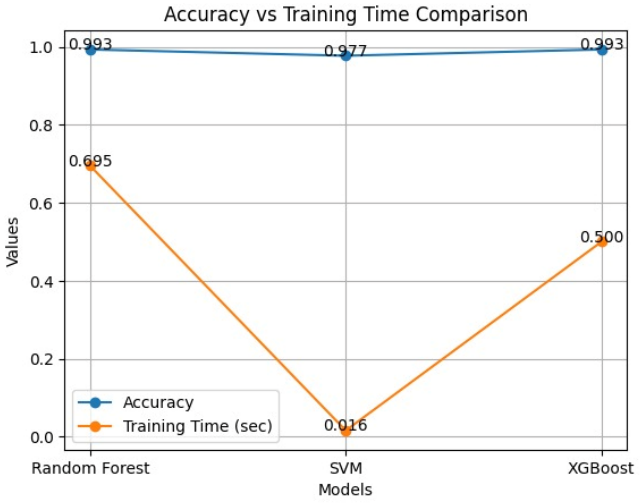


Fig.8 Accuracy Comparison Between Models.

The performance of different machine learning models used in the CropCare system was evaluated based on two important factors: prediction accuracy and training time. The models compared include Random Forest, Support Vector Machine (SVM), and XGBoost [3], [4].

The experimental results show that:

- Random Forest and XGBoost achieved the highest accuracy, both reaching approximately 99.3%, indicating strong prediction capability.
- SVM showed slightly lower accuracy (around 97.7%), but still performed reasonably well.

In terms of training time:

- SVM required the least training time (~0.016 sec), making it computationally efficient.
- Random Forest took more time (~0.695 sec) due to multiple tree construction.
- XGBoost showed moderate training time (~0.5 sec) while maintaining high accuracy.

These results indicate a trade-off between accuracy and computational efficiency. While SVM is faster,

ensemble models like Random Forest and XGBoost provide significantly better accuracy [3], [4].Based on this comparison, Random Forest and XGBoost are more suitable for deployment in the system, as they offer high accuracy with acceptable training time.

E. Market Price Prediction Model Evaluation

The performance of different regression models used for market price prediction was evaluated using R^2 Score, Root Mean Square Error (RMSE), and Mean Absolute Error (MAE) [3], [4].

The experimental results show that:

- Random Forest achieved the best performance, with the highest R^2 score (~0.99) and lowest error values (RMSE ≈ 62.09, MAE ≈ 16.92).
- XGBoost also performed well, with slightly lower R^2 (~0.98) and slightly higher error values.
- Linear Regression showed the weakest performance, with higher error values

These results demonstrate that ensemble learning methods such as Random Forest and XGBoost are more effective in capturing complex patterns in agricultural market data compared to traditional linear models [3], [4].

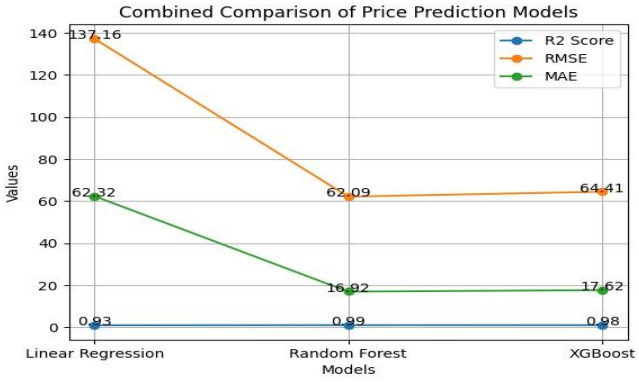


Fig. 9. Performance Comparison of Price Prediction Models

F. Disease Detection Model Performance Analysis

The performance of different classification models for plant disease detection was evaluated using Accuracy, F1 Score, and Error Rate.

The results indicate that:

- CNN achieved the highest accuracy (~92%), demonstrating strong capability in image-based classification tasks.
- SVM showed moderate performance (~57%), indicating limited effectiveness for complex image data.
- KNN performed the weakest (~30%), with higher error rate and poor generalization.

The F1 score trends also confirm that CNN provides better balance between precision and recall. Additionally, the lower error rate of CNN highlights its reliability in detecting plant diseases accurately [1], [2].

These results validate that deep learning models, especially CNN-based architectures, are more suitable for image-based disease detection compared to traditional machine learning models.

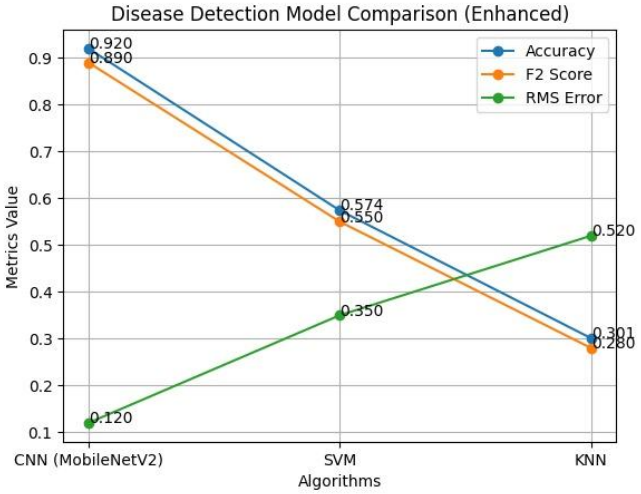


Fig.10. Performance Comparison of Price Prediction Models

G. Limitations

Although the proposed CropCare system shows strong performance across multiple modules, certain limitations are identified:

- The accuracy of predictions depends on the quality and availability of input data such as soil parameters, weather conditions, and images [2].
- The disease detection model may be affected by poor image quality, lighting conditions, or

complex backgrounds. [1]

- Market price prediction relies on historical data, which may not fully capture sudden market fluctuations or external factors [4].
- The system currently depends on internet connectivity for accessing APIs and AI assistant functionality.

6.1 System Output and Visualization Results

The developed CropCare system provides an interactive interface that allows users to access predictions and insights from multiple modules. The system outputs are visualized through a dashboard and user-friendly screens.

A. Crop Recommendation Output

The crop recommendation module generates suitable crop suggestions based on soil and environmental conditions. The output includes the recommended crop along with a confidence score**.** The interface displays the recommended crop along with prediction confidence, helping farmers select the most suitable crop [3], [9].

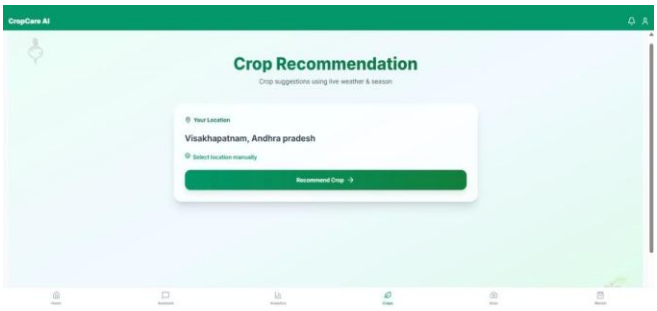


Fig. 11. Crop Recommendation Result Interface

B. Disease Detection Output

The disease detection module analyzes leaf images and identifies plant diseases using a CNN model. The system displays the detected disease along with visual confirmation, enabling early diagnosis and treatment. [1], [2].



Fig. 12. Disease Detection Output with Predicted Class

C. Market Price Prediction Output

The market price prediction module estimates future crop prices based on historical data. The output shows predicted price along with trends and supporting statistics, helping farmers make selling decisions. [7].

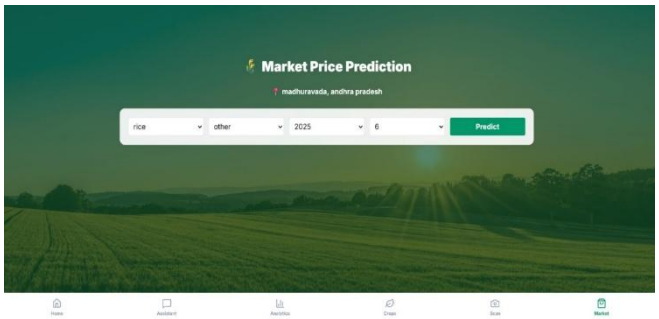


Fig. 13. Market Price Prediction Result

D. AI Assistant Interaction

The AI assistant provides real-time guidance based on user queries**.** The assistant responds to user queries and provides simplified explanations and recommendations[7], [12], [13].

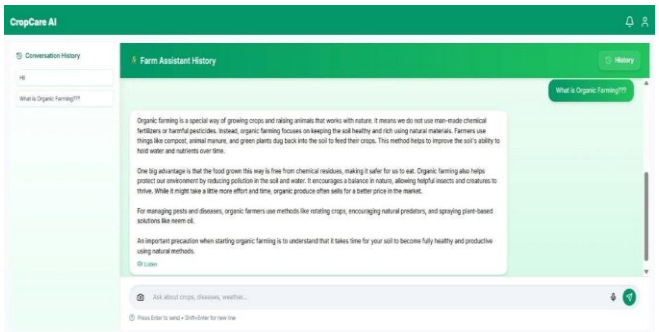


Fig. 14. AI Assistant Interaction Interface

E. Dashboard and Visualization

The dashboard integrates outputs from all modules and presents insights in graphical format. The dashboard displays combined insights such as crop suggestions, price trends, and disease results in a structured format [10].



Fig. 15. System Dashboard with Analytics and Insights

7. DISCUSSION

7.1 Strengths

The proposed CropCare system shows strong performance by integrating several smart modules into one platform. A key strength of the system is its ability to combine crop recommendations, disease detection, market price predictions, and AI support, allowing farmers to make informed decisions [9], [10]. The crop recommendation module analyzes soil and environmental conditions to suggest the best crops. The market price prediction module helps farmers select crops that are more likely to be profitable based on past trends [1], [2]. The disease detection module uses deep learning to identify plant diseases early, minimizing potential crop damage. Another important strength of the system is the AI assistant. It takes complex outputs and turns them into simple, understandable guidance. This makes the system easier to use, ensuring that even those without technical knowledge can benefit. The addition of an analysis module further improves the system by merging outputs from different modules and providing valuable insights through visualizations. This helps farmers understand trends and make better decisions instead of relying on individual predictions. Overall, the system enhances decision-making by providing clear, timely, and easy-to-understand information in a unified platform [7], [12], [13].

7.2 Scalability

The proposed system is built to be scalable and efficient for real-world agriculture. Each module crop recommendation, disease detection, market price prediction, and AI assistant works independently while still contributing to the overall function of the system. This modular design makes it easy to expand and add new features without disrupting existing components [9], [10]. The system effectively manages various types of data, including numerical data (soil, weather), image data (leaf images), and textual queries (AI assistant), allowing it to adapt to different agricultural situations. The system's efficiency relies on optimized machine learning and deep learning models [3], [4], [7]. For example, ensemble models like Random Forest and XGBoost deliver high prediction accuracy, while CNN-based models process image data effectively. The system can also be expanded to handle larger datasets, additional crops, and more geographical areas [10]. With proper deployment (such as on cloud or server-based infrastructure), the system can support multiple users at once. Although some features, like APIs and AI assistance, depend on internet connectivity, the modular design ensures that core functions can remain efficient.

8.CONCLUSION

This paper introduces CropCare, a smart farming system that uses AI to help farmers make informed and efficient choices. The system combines various functions, including crop recommendations, disease detection, market price predictions, and AI assistance, into one platform [3], [4], [7]. The crop recommendation module looks at soil and environmental conditions to suggest suitable crops. The market price prediction module analyzes historical data to forecast future prices and trends. The disease detection module uses deep learning techniques to identify plant diseases early from leaf images. Additionally, the AI assistant offers real-time guidance and simplifies system outputs into easy-to-understand recommendations [7], [12], [13]. The experimental results show that ensemble models like Random Forest and XGBoost deliver high accuracy for prediction tasks, while CNN-based models perform well in disease detection. By integrating these modules with an analysis and visualization component, the system enhances overall decision-making by bringing together various insights. The system is built to be scalable and flexible, allowing for future improvements such as integration with IoT devices, real-time data sources, and support for more crops and regions. In summary, the proposed system provides a practical and efficient solution for modern agriculture, boosting productivity, reducing losses, and supporting sustainable farming practices[10].

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